

Solution Requirements

Project Title

Virtual Agent–Driven SLA Breach Awareness & Justification System

Introduction

The Virtual Agent–Driven SLA Breach Awareness & Justification System is designed to improve IT Service Management (ITSM) by proactively monitoring Service Level Agreements (SLAs), predicting potential SLA breaches, and assisting support teams through workflow automation and AI-powered recommendations. This document defines the functional and non-functional requirements needed to successfully design, develop, deploy, and maintain the application.

Functional Requirements

The system shall provide the following functionalities:

1. User Authentication

The application shall allow authorized users to securely log in using valid credentials.

Features

- Secure login
 - User authentication
 - Session management
 - Role-based access
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2. Incident Management

The system shall enable users to manage incident records.

Features

- Create incidents
 - View incident details
 - Update incident status
 - Assign incidents to support personnel
 - Track incident lifecycle
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3. SLA Monitoring

The system shall continuously monitor Service Level Agreements associated with each incident.

Features

- Automatic SLA attachment
 - SLA countdown tracking
 - Remaining time calculation
 - SLA progress monitoring
 - Threshold detection
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4. SLA Risk Classification

The application shall classify incidents according to their SLA condition.

Categories

- SAFE
- AT RISK
- BREACHED

Classification shall be updated automatically based on the remaining SLA time.

5. Virtual Agent Assistance

The Virtual Agent shall interact with users to improve SLA awareness.

Features

- Display SLA status
 - Notify users of SLA risks
 - Explain breach reasons
 - Provide corrective guidance
 - Capture user acknowledgement
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6. AI-Based Justification

Google Gemini AI shall generate intelligent explanations for potential SLA breaches.

Features

- SLA breach explanation
- Recommended corrective actions
- Context-aware responses
- User guidance

7. Notification Management

The system shall notify users whenever SLA thresholds are reached.

Notifications

- SLA At Risk
 - SLA Breached
 - Incident updates
 - Workflow notifications
 - Email alerts
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8. Dashboard & Analytics

The application shall provide real-time dashboards.

Dashboard Information

- Total incidents
 - SAFE incidents
 - AT RISK incidents
 - BREACHED incidents
 - SLA compliance
 - Agent performance
 - Resolution trends
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9. Reporting

The application shall generate reports for administrators and managers.

Reports include:

- SLA compliance reports
 - Incident reports
 - Breach justification reports
 - Agent performance reports
 - SLA trend analysis
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Non-Functional Requirements

The application shall satisfy the following quality attributes.

Performance

The system should:

- Monitor incidents in real time.
 - Generate notifications with minimal delay.
 - Display dashboards efficiently.
 - Handle multiple users simultaneously.
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Reliability

The application should:

- Maintain accurate SLA calculations.
 - Provide consistent workflow execution.
 - Ensure reliable AI response generation.
 - Prevent data inconsistencies.
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Security

The application shall implement:

- Secure authentication
 - Role-Based Access Control (RBAC)
 - Protected incident records
 - Secure API communication
 - User activity logging
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Availability

The system should remain available during normal business operations with minimal downtime.

Scalability

The application should support:

- Increasing numbers of incidents.
- Additional support teams.

- Larger organizational deployments.
 - Future feature enhancements.
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Usability

The system should provide:

- Simple user interface.
 - Easy navigation.
 - Clear SLA indicators.
 - Interactive dashboards.
 - User-friendly Virtual Agent conversations.
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Maintainability

The solution should be easy to:

- Update workflows.
 - Modify business rules.
 - Expand dashboard functionality.
 - Integrate additional AI capabilities.
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Hardware Requirements

Minimum hardware requirements include:

- Dual-Core Processor or higher
 - 4 GB RAM (Minimum)
 - Stable Internet Connection
 - Modern Web Browser
 - Desktop or Laptop Computer
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Software Requirements

Component	Technology
Backend	Python (Flask)
Frontend	HTML, CSS, JavaScript

Component	Technology
Charts	Chart.js
Database	SQLite
AI Integration	Google Gemini AI
Workflow Platform	ServiceNow
Automation	Flow Designer
Operating System	Windows / Linux
Browser	Google Chrome / Microsoft Edge

User Roles

The application supports three primary user roles.

Support Agent

Responsibilities:

- Monitor incidents
 - Respond to Virtual Agent alerts
 - Update incidents
 - Provide SLA breach justifications
 - Acknowledge SLA risks
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Team Lead / Manager

Responsibilities:

- Monitor team performance
 - Review SLA reports
 - Analyze breach trends
 - Track SLA compliance
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Administrator

Responsibilities:

- Manage users
- Configure workflows

- Maintain dashboards
 - Generate reports
 - Monitor overall system performance
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Assumptions

The following assumptions were considered during development:

- Users possess valid login credentials.
 - Every incident is associated with an SLA.
 - Network connectivity is available.
 - Google Gemini AI services are accessible.
 - Email notifications are configured correctly.
 - Users interact with the Virtual Agent through the Service Portal.
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Constraints

The system is subject to the following constraints:

- Internet connectivity is required.
 - AI responses depend on external API availability.
 - Workflow execution depends on ServiceNow platform services.
 - Dashboard accuracy depends on the latest incident data.
 - Performance may vary with increasing incident volumes.
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Conclusion

The Solution Requirements document defines the essential functional and non-functional requirements for the Virtual Agent–Driven SLA Breach Awareness & Justification System. These requirements establish a strong foundation for designing an intelligent, secure, and scalable application capable of proactively managing SLA risks, improving operational efficiency, and enhancing IT service delivery through workflow automation and AI-assisted decision-making.